

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30 June 2026
Team ID	[Team ID]
Project Name	Customer Care Registry
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the Customer Care Registry, derived directly from the implemented modules and API endpoints of the application.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	- Registration with name, email, phone, address (Customer) - Login with email & password, JWT-based session - Role-based access: Super Admin, Support Manager, Support Agent, Customer
FR-2	Customer Management	- Add / edit / view customer records (name, contact, address, type: Enterprise/Standard/VIP) - Deactivate or delete a customer record (Super Admin only) - View complete interaction & complaint history per customer
FR-3	Complaint / Ticket Management	- Raise a complaint with title, description, category, priority, and attachments - Auto-generate a unique ticket ID (e.g. TKT-2026-0001) - Assign ticket to a support agent and track status through its lifecycle (New → Assigned → In Progress → Waiting for Customer → Escalated → Resolved → Closed / Rejected) - Maintain a timeline of all actions taken on a ticket
FR-4	AI-Assisted Categorization	- Automatically suggest category, priority, and department from complaint text using the Gemini API - Fall back to keyword-based heuristics when the AI service is unavailable - Generate AI-assisted reply suggestions for agents
FR-5	Interaction Logging	- Log customer touchpoints: Phone Call, Email, Chat, Visit, WhatsApp, Video Call - Record summary, notes, attachments, and optional follow-up date
FR-6	Feedback Management	- Collect rating (1–5), comment, and resolution satisfaction after a ticket is resolved - Derive sentiment (Positive / Neutral / Negative) from feedback
FR-7	Knowledge Base	- Create and publish FAQ / Solution / Article entries - Search articles by keyword and track view counts
FR-8	Analytics Dashboard	- Display metrics: total/active customers, complaints today, pending & resolved counts, average resolution time, average satisfaction score - Show charts: top complaint categories, monthly filed vs resolved trends, department performance, agent performance
FR-9	Audit Logging	- Record every significant user action (who, what, when) for accountability - Restrict viewing of audit logs to Super Admin and Support Manager
FR-10	System Settings	- Configure SLA hours per priority level (Low, Medium, High, Critical) - Enable / disable automatic escalation of overdue tickets

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, role-based dashboard UI (React + Tailwind) with a consistent sidebar/header layout so agents, managers, and customers only see relevant modules for their role.
NFR-2	Security	JWT-based authentication, role-based access control (RBAC) middleware on every API route, and audit logging of sensitive actions (user, customer, and settings changes).
NFR-3	Reliability	AI categorization has a deterministic keyword-based fallback so complaint intake keeps working even if the Gemini API is offline or unconfigured.
NFR-4	Performance	REST endpoints return filtered, paginated-ready JSON; analytics are computed on demand from the data store to keep dashboard responses fast for typical data volumes.
NFR-5	Availability	Stateless Express API (server.ts) can be restarted or redeployed independently of the React front end; static build output can be served from any Node host.
NFR-6	Scalability	Layered architecture (UI → REST API → data layer) allows the current JSON file store (server/db.ts) to be swapped for a scalable database (e.g. MongoDB/PostgreSQL) without changing the API contract or front end.